

The 2022 Oil Price Shock and Fossil Fuel Subsidies in Latin America and the Caribbean: A Difference-in-Differences Analysis

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Abstract

The rise in oil prices that began with the post-COVID-19 recovery in 2021 and intensified after the Russian invasion of Ukraine in 2022 was a common shock to every country in Latin America and the Caribbean. I use it to estimate whether explicit fossil fuel subsidies responded differently across countries depending on their net oil trade position, defined by a rule fixed before estimation: a country is a net oil exporter if its 2015–2019 net trade in crude oil and refined products (UN Comtrade) is positive. With a panel of 34 countries over 2015–2023 from the IMF Fossil Fuel Subsidies Database and a two-way fixed effects difference-in-differences design, I estimate that the explicit subsidy of the five net oil exporters rose by 1.32 percentage points (pp) of GDP more than that of net importers in 2022 (clustered SE 1.03, $p = 0.21$; 95% CI $[-0.79, 3.42]$). The estimate is not statistically significant, and it depends heavily on Venezuela: without it, the differential falls to 0.28 pp. Point estimates are positive under every binary definition of exporters considered (0.92 to 2.06 pp), so the sign is robust but significance is not. A continuous measure of net oil exposure, pre-specified as a sensitivity check, points to a dose–response pattern (0.223 pp of GDP per 1 pp of GDP of net oil exports), but it is heavily leveraged by Venezuela. A pre-trend test does not reject parallel trends ($p = 0.26$), although it has little power. The evidence is suggestive, not conclusive. A descriptive subsidy–debt matrix shows that fiscal pressure is not confined to exporters: among the 26 countries with non-negligible explicit subsidies, eight, six of them net importers, combine above-median subsidies with above-median public debt. I argue for a reform sequence differentiated by fiscal space, with targeted compensation replacing universal price subsidies.

JEL codes: H23, H62, Q48, C23.

Keywords: fossil fuel subsidies, oil price shock, difference-in-differences, fiscal space, Latin America and the Caribbean.

1 Introduction

In 2022, the annual average price of Brent crude rose from USD 70.9 to USD 100.9 per barrel, an increase of 42%. The increase began with the post-COVID-19 recovery in 2021 and intensified after the Russian invasion of Ukraine in February 2022, an event that was unexpected, global in scope, and unrelated to the subsidy decisions of any government in Latin America and the Caribbean (LAC). Worldwide, fossil fuel subsidies reached USD 7 trillion, or 7.1% of world GDP, in 2022, and explicit subsidies more than doubled relative to 2020 (Black et al. 2023). Governments that hold domestic fuel prices below supply cost absorb international price increases in their budgets, so a shock of this size is also a fiscal event.

This paper asks how explicit fossil fuel subsidies in LAC responded to the 2022 shock and what that response implies for fiscal and subsidy policy. The starting point is that the same price increase operates through different channels depending on a country's net oil trade position. For a net oil exporter, a higher Brent price raises oil revenue, which makes it easier to keep the domestic price low: the subsidy expands, and there is rent to finance it. For a net importer, the higher price raises the cost of supply; if the consumer price is frozen, the subsidy also grows, but without additional revenue to back it. Because exporters have the rent to finance a larger price gap, I expect their explicit subsidy to rise more than importers', and more so the larger their net oil exposure (Section 4.1). Exporter status is also correlated with the degree of domestic price regulation, which, in turn, makes the explicit subsidy respond more to international prices. The design identifies a differential response by exporter status; it cannot separate the oil-rent channel from differences in pricing regimes.

That subsidies rose in 2022 is a descriptive fact, not an estimate of the shock's effect. I cannot observe what exporters' subsidies would have been without the shock, because every country in the region faced the same price increase. Instead, I compare how subsidies moved in net exporters with how they moved in net importers, which went through the same years, including the post-COVID-19 rebound and the 2022 inflation, and were exposed to the same price but not to the oil-rent channel. Netting out the change common to both groups yields a difference-in-differences (DiD) design, identified from the interaction between the shock year and exporter status. The resulting estimate is a *differential* effect: it measures how much more exporters' subsidies rose than importers', not the total effect of the shock on exporters.

The data combine the IMF Fossil Fuel Subsidies Database, which distinguishes between explicit and implicit subsidies (International Monetary Fund 2023a; Black et al. 2023), the EIA Brent price series (U.S. Energy Information Administration 2023), oil trade flows from UN Comtrade (United Nations 2026), and fiscal indicators from the IMF World Economic Outlook (International Monetary Fund 2023b). The panel covers 34 LAC countries, five of them net oil exporters, over 2015–2023 (306 country-years). Exporter status follows a data-based rule that I fixed before estimating any model; an earlier version of this analysis used a hand-coded list of exporters, which I now report only as a sensitivity check.

There are five main findings. First, the two-way fixed effects (TWFE) estimate implies that the explicit subsidy of net exporters rose by 1.32 pp of GDP more than that of net importers in 2022, with a clustered standard error of 1.03 ($p = 0.21$; 95% CI $[-0.79, 3.42]$). The point estimate is about 35% of the unweighted

pre-shock mean of exporters (3.74% of GDP), but it is not statistically significant, and with only five treated countries, conventional clustered standard errors are, if anything, too small. Second, the estimate depends heavily on one country: without Venezuela, it falls to 0.28 pp ($p = 0.54$), and dropping each exporter in turn gives estimates between 0.28 and 1.81 pp. Third, the sign is robust to how exporters are defined, but significance is not: across seven binary definitions, the point estimate lies between 0.92 and 2.06 pp, and none is significant at the 5% level; the earlier hand-coded list gives 1.79 pp ($p = 0.06$). Fourth, a continuous measure of net oil exposure points to a dose–response pattern: each additional percentage point of GDP in net oil exports is associated with a 0.223 pp larger differential ($p < 0.001$). This evidence is complementary and heavily leveraged by Venezuela, whose net oil exports equal 19.0% of GDP; without it, the slope falls to 0.130 ($p = 0.010$). Fifth, the timing is consistent with a price channel but does not establish it: the event-study coefficient peaks in 2022 (+1.77 pp) and falls to +0.44 pp in 2023, when Brent declined; a joint test does not reject that the six pre-shock coefficients are all zero ($p = 0.26$), but it has little power; and the implicit subsidy, a falsification outcome, shows no differential (−0.09 pp, $p = 0.83$). Taken together, the results are suggestive of a larger subsidy response among net oil exporters, not conclusive.

The paper provides a transparent estimate, based on country-level data and a pre-specified treatment rule, of how a common oil price shock propagated differently across countries’ budgets in a region with very heterogeneous exposure, and it links that estimate to a descriptive map of fiscal space. The policy analysis shows that pressure is not limited to exporters: several net importers combine high subsidies with high debt. Section 2 describes the context, data, and treatment definition; Section 3 presents descriptive evidence; Section 4 sets out a conceptual framework and the empirical strategy; Section 5 reports the results; Section 6 examines the identifying assumptions; Section 7 reports robustness checks, including the sensitivity to the definition of exporters, and discusses inference; Section 8 discusses budgetary implications and policy; and Section 9 concludes.

2 Context and data

2.1 The 2022 shock and the subsidy measures

The main data source is the IMF Fossil Fuel Subsidies Database (International Monetary Fund 2023a), documented in Black et al. (2023). It reports subsidies by country, year, fuel, and end use under the database’s baseline scenario, and it separates two concepts. The *explicit* subsidy is the gap between the price paid by consumers and the cost of supply, multiplied by consumption; it is a budgetary or quasi-fiscal cost, and it responds mechanically to the international price when domestic prices are regulated. The *implicit* subsidy captures environmental and health externalities that are not priced, as well as forgone consumption taxes. It is dominated by slow-moving structural components such as consumption volumes and damage parameters.

The outcome variable of the analysis is the explicit subsidy as a percentage of GDP, because it is the component that should react to an international price shock in the short run. The implicit subsidy serves as a falsification outcome. It is an imperfect one: it is measured relative to the higher of the retail price and the supply cost, and it includes forgone value-added tax, so the two measures are constructed jointly and a

price shock can move the implicit subsidy mechanically in either direction. I therefore read it as supporting evidence rather than as a formal test.

The shock variable is the annual average of the Europe Brent spot price (U.S. Energy Information Administration 2023), a series the IMF database does not include. In the 2015–2023 window, the 2022 increase of USD 30.1 per barrel is the largest year-on-year rise in dollar terms, and 2022 has the highest annual average. In percentage terms, the 2021 rebound from the pandemic trough was larger (+69%), which is one reason the design compares groups rather than years. In 2023 the price fell back to USD 82.5.

2.2 Fiscal variables

Fiscal indicators are sourced from the IMF World Economic Outlook, accessed via the IMF DataMapper API (International Monetary Fund 2023b): gross general government debt, the general government fiscal balance, and general government revenue, all expressed as a percent of GDP. I use the WEO rather than the World Bank because the policy analysis compares debt with subsidies country by country, which requires full coverage. The WEO series have no missing values for the 34 countries over 2015–2023. The corresponding World Bank series are missing for 45% to 79% of country-years depending on the indicator, with the gaps concentrated in the Caribbean, which would have reduced the fiscal comparison to a handful of cases.

The fiscal variables do not enter the DiD model. Debt, the fiscal balance, and revenue are themselves affected by the shock and by the subsidy; they are outcomes, not predetermined covariates, and conditioning on them would introduce bias. They are used only in the policy analysis of Section 8.

2.3 Sample

The panel contains all 34 LAC countries covered by the IMF database over 2015–2023, giving 306 country-years. The window starts in 2015, the first year extracted from the database, and includes seven pre-shock years (2015–2021) to assess parallel trends, the shock year (2022), and a reversal year (2023). Coverage of the explicit subsidy is nearly complete: the only missing values are for Puerto Rico in 2015–2017. Coverage of the implicit subsidy is less complete: it is missing for all years for Aruba, Puerto Rico, St. Kitts and Nevis, and Suriname, and for most years for Antigua and Barbuda and Dominica, which matters for the falsification regression in Section 5.

2.4 Treatment definition

Treatment status is defined by the net oil trade position, not by whether a country produces oil. What determines whether a government gains revenue when Brent rises is the net balance of oil trade. I fixed the classification rule before estimating any model: a country is a net oil exporter if its average net trade in crude oil and refined petroleum products (HS headings 2709 and 2710; exports minus imports in current USD, partner World) over 2015–2019 is positive, and a net oil importer otherwise. The trade data come from UN Comtrade (United Nations 2026). Venezuela has not reported to Comtrade in any year of the window, so its position is rebuilt from mirror data, that is, from its partners' reported imports from and exports to Venezuela.

Three countries reported only some years of the window (Haiti and St. Kitts and Nevis three, Dominica four); their averages use the years they reported, and all three are clearly net importers. The measure is taken before the shock and held fixed over time, so treatment status cannot respond to the 2022 price increase or to the subsidy response itself. The 2015–2019 window also ends before the pandemic and before Guyana’s oil exports: production started in December 2019 and exports in 2020. Guyana was a net importer over the window and is classified as such; it appears as an exporter only in the 2019–2021 variant and in the earlier hand-coded list, both reported in Section 7.

The rule yields five net exporters, Brazil, Colombia, Ecuador, Trinidad and Tobago, and Venezuela; the remaining 29 countries are net importers. Argentina illustrates why production is not the criterion: it exports some crude, but imports more refined products than it exports, so a higher Brent price raises the cost of supply more than it raises oil revenue. Two cases need comment. Puerto Rico does not appear in Comtrade, which records its trade under the United States, so the rule cannot be applied to it; I classify it as a net importer based on EIA data, which report that it has no crude oil production and no operating refinery (U.S. Energy Information Administration 2026). Trinidad and Tobago is a knife-edge case: its average net oil trade is only USD 8 million a year, negative in 2015 and 2016 and positive in 2017–2019, so a small change in the window could reclassify it. Section 7 reports the estimates without each of these two countries.

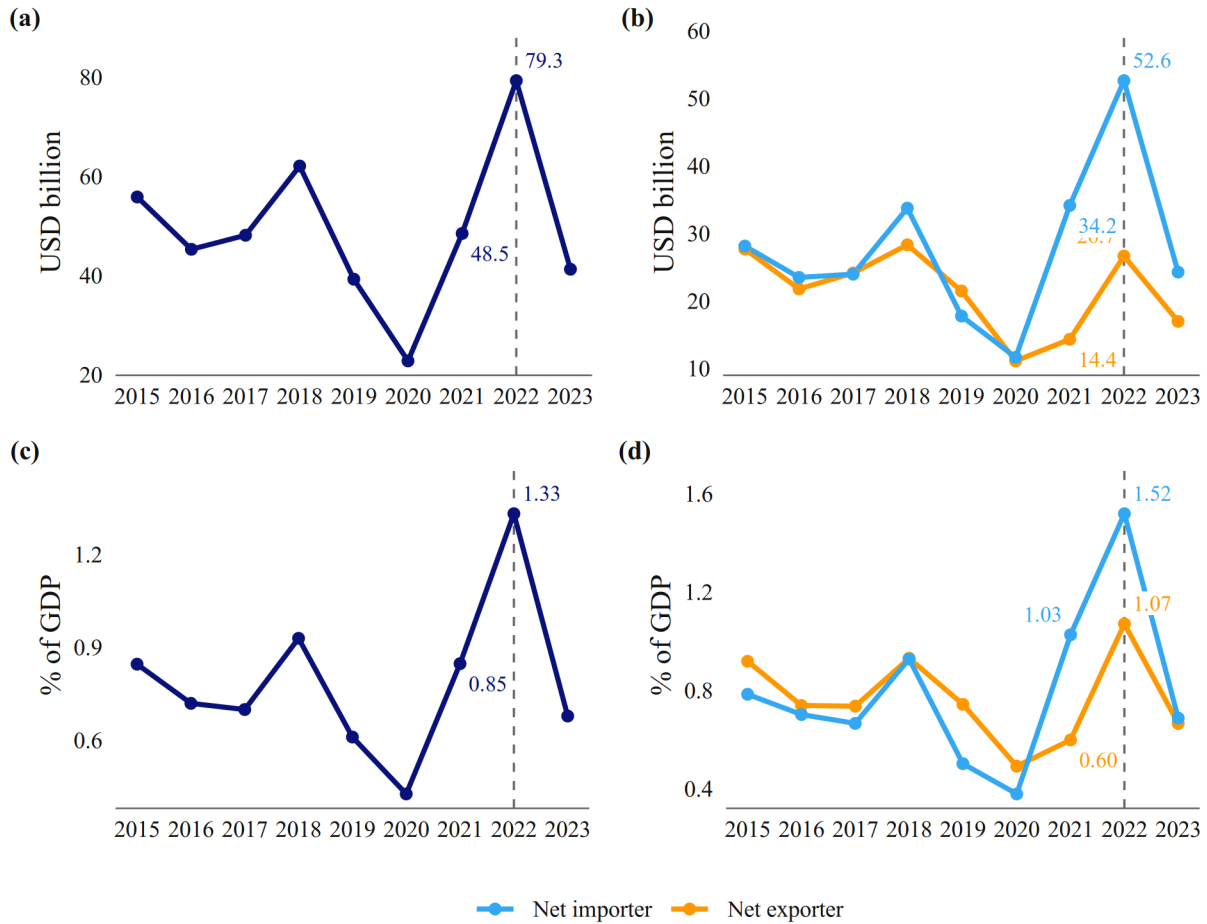
The data-based rule differs from the hand-coded list of seven exporters used in an earlier version of this analysis in four countries. Mexico and Bolivia become net importers. Mexico’s net crude exports (USD 20.6 billion a year) are outweighed by its net imports of refined products (USD 22.3 billion). Bolivia’s hydrocarbon exports are mainly natural gas (HS 2711), which the rule does not count, and its refined-product imports exceed its crude exports. Guyana becomes a net importer for the timing reason given above. Brazil becomes a net exporter, because its net crude exports (USD 12.9 billion a year) exceed its net imports of refined products (USD 8.4 billion). I report the hand-coded list only as a sensitivity check, although it yields a smaller *p*-value than the main rule; a classification chosen after seeing the results would invalidate the inference. Table 2 in Section 3 lists the classification with each country’s explicit subsidy by year.

The treated group is small, and this shapes the rest of the analysis. With five exporters, standard errors are large, confidence intervals are wide, and conventional inference is fragile (see Section 7).

3 Descriptive evidence

Figure 1 plots the explicit subsidy by group and year, in USD billion (panels a and b) and as a share of GDP (panels c and d). Between 2021 and 2022, the LAC total rose from USD 48.5 billion to USD 79.3 billion (+63.5%), or from 0.85% to 1.33% of GDP. In these aggregates, the increase was not concentrated among exporters. Among net exporters, the subsidy went from 0.60% to 1.07% of GDP (+0.47 pp), while among net importers it went from 1.03% to 1.52% (+0.49 pp; changes are computed from unrounded values). In dollar terms, exporters account for USD 12.3 billion of the USD 30.8 billion increase. The series also shows the volatility of the pre-shock years, including the 2018 peak and the 2020 collapse, both of which reappear in the event study.

Figure 1. Explicit fossil fuel subsidies by exposure group, 2015–2023



Notes: Explicit fossil fuel subsidy (gap between the supply cost and the consumer price), annual group totals: USD billion in panels (a)–(b) and percentage of the group’s aggregate GDP in panels (c)–(d). The dashed line marks the 2022 shock, when the LAC total rose from USD 48.5 billion to USD 79.3 billion (+63.5%). Countries are classified by net oil trade position (average 2015–2019 net exports of crude oil and refined products, UN Comtrade), not by production. Net oil exporters (five): Brazil, Colombia, Ecuador, Trinidad and Tobago, and Venezuela; the other 29 countries are net oil importers. N = 34 countries, 2015–2023. *Source:* IMF Fossil Fuel Subsidies Database; UN Comtrade.

Table 1 reports the underlying numbers for the explicit, implicit, and total subsidy. The aggregates in Figure 1 and Table 1 are group totals divided by group GDP, so large economies weigh more: Brazil, the largest exporter economy, has an explicit subsidy close to zero, and Mexico, the largest importer economy, saw its subsidy rise in 2022. The regressions in Section 5 weight every country equally, so a small economy with a large subsidy, such as Venezuela, counts as much as Brazil. The two approaches, therefore, give different pictures, and the differential estimated in Section 5 comes from the equal-weighted comparison. The implicit subsidy also rose in 2022 for both groups: from 4.20% to 4.32% of GDP among exporters and from 3.72% to 5.04% of GDP among importers. That increase is consistent with the price-sensitive tax component of the implicit measure and helps explain why the falsification outcome must be read with caution; the regression in

Section 5 asks whether the implicit subsidy rose differentially among exporters once country and year effects are removed.

Table 1. Explicit, implicit and total subsidies by exposure group (USD billion and % of GDP), 2015–2023

Variable	2015	2016	2017	2018	2019	2020	2021	2022	2023
Panel A. LAC total									
Explicit (USD billion)	55.9	45.3	48.2	62.1	39.3	22.8	48.5	79.3	41.3
Implicit (USD billion)	192.4	197.5	197.4	204.9	204.1	158.9	224.3	282.2	231.6
Total (USD billion)	265.3	258.7	257.2	275.5	252.4	194.0	288.6	378.1	290.0
Explicit (% of GDP)	0.85	0.72	0.70	0.93	0.61	0.43	0.85	1.33	0.68
Implicit (% of GDP)	2.91	3.14	2.87	3.07	3.17	2.97	3.92	4.74	3.81
Total (% of GDP)	4.02	4.11	3.74	4.12	3.92	3.63	5.04	6.35	4.77
N (countries)	34	34	34	34	34	34	34	34	34
Panel B. Net oil exporters									
Explicit (USD billion)	27.7	21.8	24.2	28.3	21.5	11.2	14.4	26.7	17.0
Implicit (USD billion)	78.5	69.9	73.3	77.1	79.7	69.2	100.7	107.5	96.1
Total (USD billion)	111.6	96.7	102.7	109.9	105.2	83.1	117.6	136.8	115.9
Explicit (% of GDP)	0.92	0.74	0.74	0.93	0.74	0.49	0.60	1.07	0.67
Implicit (% of GDP)	2.60	2.37	2.24	2.53	2.76	3.05	4.20	4.32	3.76
Total (% of GDP)	3.70	3.28	3.13	3.61	3.64	3.67	4.91	5.50	4.54
N (countries)	5	5	5	5	5	5	5	5	5
Panel C. Net oil importers									
Explicit (USD billion)	28.2	23.5	24.0	33.8	17.8	11.7	34.2	52.6	24.3
Implicit (USD billion)	113.9	127.6	124.0	127.8	124.4	89.7	123.6	174.6	135.6
Total (USD billion)	153.8	161.9	154.6	165.6	147.2	110.9	171.0	241.3	174.1
Explicit (% of GDP)	0.78	0.70	0.67	0.93	0.50	0.38	1.03	1.52	0.69
Implicit (% of GDP)	3.18	3.81	3.45	3.51	3.51	2.92	3.72	5.04	3.84
Total (% of GDP)	4.29	4.84	4.29	4.55	4.16	3.60	5.14	6.97	4.93
N (countries)	29	29	29	29	29	29	29	29	29

Notes: Group totals by year, in USD billion and as a percentage of the group's aggregate GDP. Explicit: gap between the supply cost and the consumer price. Implicit: uninternalized externalities and forgone VAT, driven mainly by volumes and damage parameters but not independent of the price (its forgone-VAT part scales with it); not estimated for every country-year. Total: the IMF's own aggregate, which need not equal explicit plus implicit. Countries are classified by net oil trade position (average 2015–2019 net exports of crude oil and refined products, UN Comtrade), not by production. Net oil exporters (five): Brazil, Colombia, Ecuador, Trinidad and Tobago, and Venezuela; the other 29 countries are net oil importers. *Source:* IMF Fossil Fuel Subsidies Database; UN Comtrade.

Table 2 reports the explicit subsidy of each country by year, grouped by exposure, together with its change in 2022 relative to the 2015–2021 average. The levels are highly heterogeneous within both groups, ranging from Venezuela, whose subsidy ranged from 8.3% to 18.1% of GDP over 2015–2023, to the eight countries, including Brazil, whose 2022 subsidy was at most 0.05% of GDP.

Figure 2 connects the shock and the outcome. When the international price rises and is not passed through to consumers, the gap between supply cost and the regulated price widens, and the explicit subsidy grows almost mechanically. Over 2015–2023, the Brent price and the aggregate explicit subsidy of LAC (in % of GDP) have a correlation of 0.75, with a joint peak in 2022. This correlation describes the link; it does not identify an effect. It pools exporters, where a higher price inflates the rent that finances the subsidy, with

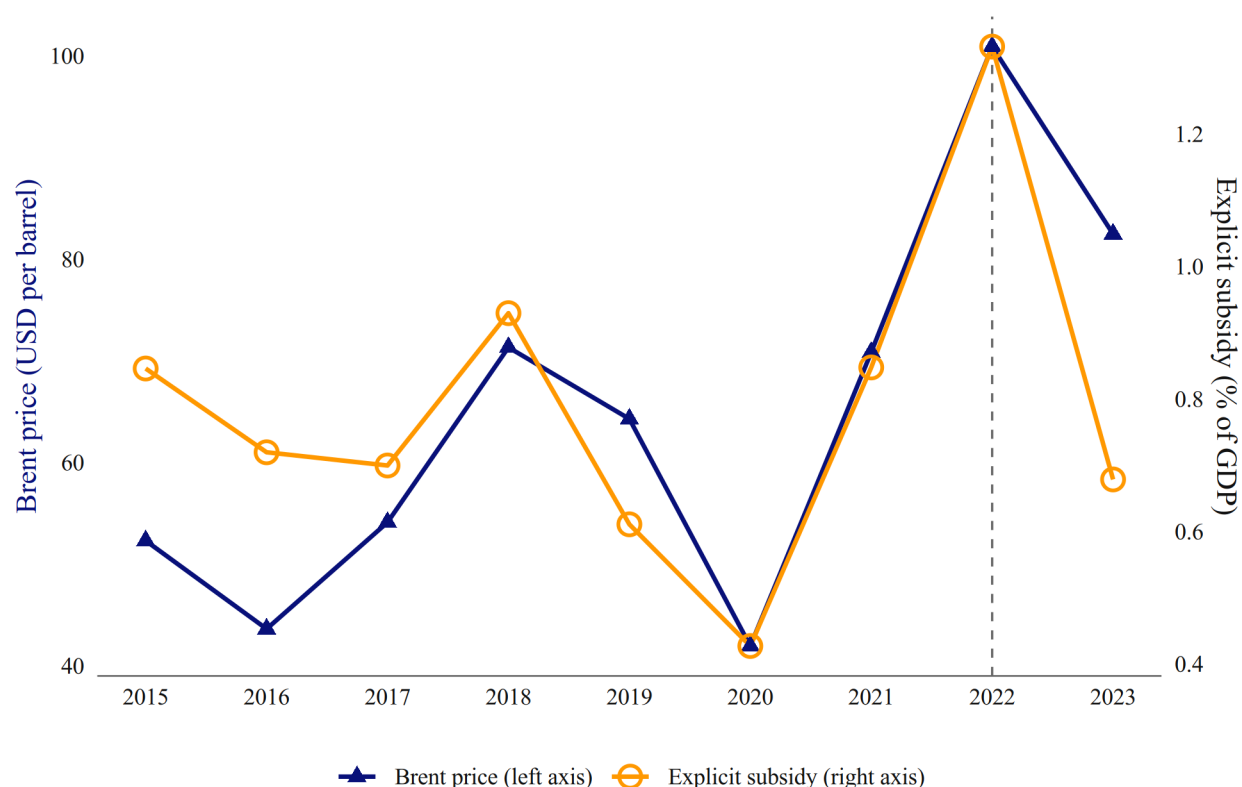
Table 2. Classification of the 34 countries by net oil exposure and annual explicit subsidy (% of GDP)

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	Change (pp)
Panel A. Net oil exporters										
Venezuela	12.35	9.99	11.37	15.61	14.74	8.33	11.25	18.06	13.00	6.12
Colombia	0.16	0.13	0.09	0.13	0.20	0.19	0.81	2.17	0.51	1.93
Ecuador	4.07	4.08	4.16	4.77	4.15	3.80	2.95	5.02	4.47	1.02
Trinidad and Tobago	3.04	2.38	2.29	2.54	1.85	2.41	2.66	3.19	2.07	0.74
Brazil	0.04	0.05	0.04	0.05	0.08	0.07	0.07	0.04	0.04	-0.02
N (countries) = 5										
Panel B. Net oil importers										
Bolivia	2.72	1.40	1.94	3.98	2.72	0.91	4.80	8.02	5.03	5.38
Suriname	3.32	3.31	2.65	2.14	2.94	3.58	4.70	8.49	6.45	5.25
Argentina	1.79	1.02	0.75	1.42	1.33	1.09	2.95	3.65	2.33	2.17
Haiti	0.10	0.65	0.94	0.78	1.43	0.48	1.89	2.32	1.66	1.42
Dominican Republic	0.53	0.53	0.34	0.67	0.79	0.81	2.01	2.06	1.50	1.25
Mexico	0.64	0.95	1.02	1.31	0.39	0.31	0.92	1.64	0.34	0.85
Belize	0.05	0.04	0.09	0.16	1.05	0.12	0.42	0.85	0.82	0.58
El Salvador	0.55	0.64	0.38	0.59	0.75	0.28	0.58	1.04	1.11	0.50
Nicaragua	0.01	0.00	0.08	0.07	0.11	0.03	0.21	0.43	0.34	0.36
Jamaica	0.62	0.85	1.36	2.52	2.22	0.44	0.76	1.59	1.17	0.34
Honduras	0.09	0.24	0.38	0.24	0.54	0.46	0.33	0.60	0.12	0.28
Guatemala	0.14	0.07	0.08	0.14	0.26	0.15	0.19	0.42	0.46	0.27
Antigua and Barbuda	0.00	0.00	0.00	0.12	0.03	0.00	0.37	0.33	0.39	0.25
Peru	0.08	0.13	0.02	0.00	0.00	0.00	0.00	0.15	0.11	0.12
Dominica	0.71	0.41	0.33	0.19	0.24	0.24	0.46	0.45	0.30	0.08
Uruguay	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.05	0.00	0.04
Costa Rica	0.07	0.07	0.07	0.09	0.13	0.07	0.14	0.13	0.10	0.04
Panama	0.11	0.08	0.05	0.05	0.06	0.06	0.06	0.07	0.06	0.00
Paraguay	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.00
Aruba	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Puerto Rico	n.a.	n.a.	n.a.	0.00	0.01	0.00	0.00	0.00	0.00	0.00
Bahamas	0.01	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Barbados	0.33	0.33	0.36	0.40	0.37	0.32	0.38	0.35	0.29	-0.01
St. Vincent and the Grenadines	0.11	0.09	0.07	0.07	0.04	0.03	0.00	0.00	0.00	-0.06
St. Kitts and Nevis	0.14	0.10	0.09	0.08	0.06	0.03	0.00	0.00	0.00	-0.07
Chile	0.21	0.10	0.12	0.13	0.15	0.15	0.16	0.07	0.08	-0.07
St. Lucia	0.36	0.35	0.29	0.31	0.30	0.34	0.26	0.22	0.21	-0.09
Grenada	1.24	1.11	1.09	0.38	0.37	0.35	0.33	0.43	0.14	-0.26
Guyana	0.18	1.00	1.16	1.28	1.36	1.22	1.05	0.77	0.41	-0.27
N (countries) = 29										

Notes: Explicit fossil fuel subsidy as a percentage of GDP, by country and year. Change (pp): 2022 value minus the 2015–2021 average, in percentage points; countries are sorted by this change within each panel. A value of 0.00 indicates a subsidy below 0.005% of GDP; n.a.: not available (Puerto Rico 2015–2017). Countries are classified by net oil trade position (average 2015–2019 net exports of crude oil and refined products, UN Comtrade), not by production (five net oil exporters and 29 net oil importers). Puerto Rico, absent from UN Comtrade, is classified as a net oil importer from U.S. Energy Information Administration data. *Source:* IMF Fossil Fuel Subsidies Database; UN Comtrade.

importers, where it raises the cost of supply, and it attributes to the price everything else that moved in the same years.

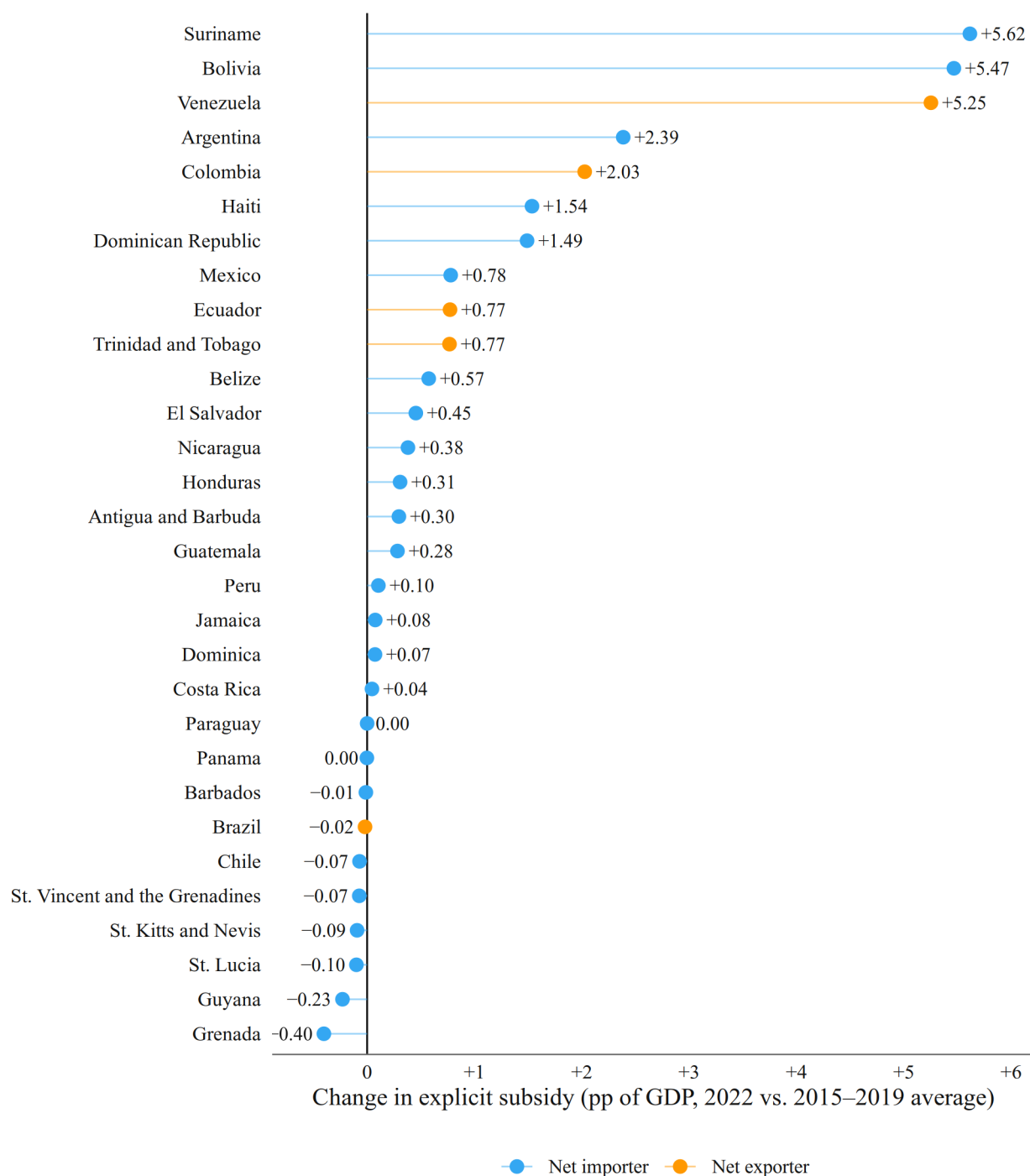
Figure 2. Brent price and aggregate explicit subsidy in LAC, 2015–2023



Notes: Brent: annual average Europe Brent spot price (USD per barrel; left axis, triangles), the shock variable. Explicit subsidy: gap between the supply cost and the consumer price, LAC total as a percentage of regional GDP (right axis, circles). The dashed line marks the 2022 shock. The correlation between the two series over 2015–2023 is 0.75. The two axes have different scales, so the figure shows co-movement, not comparable levels. *Source:* IMF Fossil Fuel Subsidies Database; U.S. Energy Information Administration (EIA).

Figure 3 shows the change in the explicit subsidy by country, comparing 2022 with the 2015–2019 average, for the 30 countries whose subsidy exceeds 0.05% of GDP in at least one year. Three pre-shock baselines appear in the paper, and each serves a different purpose. Figure 3 uses 2015–2019 to exclude the pandemic years from the country-level comparison; Table 2 and the DiD model use the full 2015–2021 pre-period; and Table 6 reports the observed 2021–2022 change. The same three countries have the largest increases under the 2015–2019 and 2015–2021 baselines. The subsidy rose in 21 of the 30 countries (in Paraguay by less than 0.01 pp). The three largest increases are in Suriname (+5.62 pp), Bolivia (+5.47 pp), and Venezuela (+5.25 pp). Two of the three, Suriname and Bolivia, are net importers, which already signals that fiscal pressure is not exclusive to exporters. Argentina, also a net importer, ranks fourth (+2.39 pp), above four of the five exporters. At the other end, Brazil, a net exporter whose explicit subsidy was close to zero throughout, recorded a marginal decrease (–0.02 pp). The DiD estimate is an average differential between groups, not a description of each country’s response, and this heterogeneity motivates two later choices: equal country weights in the model, and a country-by-country policy analysis in Section 8.

Figure 3. Change in the explicit subsidy by country, 2022 versus the 2015–2019 average



Notes: Change in the explicit fossil fuel subsidy between the 2015–2019 average and 2022, in percentage points of GDP; the base period excludes the pandemic years 2020–2021. The subsidy rose in 21 of the 30 countries. The measure is descriptive, not a causal estimate. Countries with an explicit subsidy above 0.05% of GDP in at least one year of 2015–2023 (five net oil exporters and 25 net oil importers, classified by net oil trade position, not by production). *Source:* IMF Fossil Fuel Subsidies Database.

4 Empirical strategy

4.1 Conceptual framework

A simple accounting identity organizes the predictions. The explicit subsidy of country i is the gap between the supply cost and the regulated consumer price, times the quantity consumed:

$$ES_i = [c(P) - \bar{p}_i] Q_i,$$

where $c(P)$ is the supply cost, increasing in the Brent price P , $\bar{p}_i$ is the regulated domestic price, and Q_i is consumption. If $\bar{p}_i$ is sticky, an increase dP raises $c(P)$ in every country, and the explicit subsidy rises everywhere, in importers as well as exporters. This common response is what the year effects of Section 4.3 absorb.

Exporters differ in one respect: they also receive oil rents $R_i(P)$, which rise with the price in proportion to net oil exports N_i . The windfall relaxes the government's budget constraint and makes it cheaper to hold $\bar{p}_i$ fixed rather than pass the price increase through to consumers. If π_i denotes the share of the cost increase passed through, the change in the subsidy is approximately

$$dES_i \approx (1 - \pi_i) c'(P) Q_i dP, \quad \pi_i = \pi(N_i), \quad \pi'(N_i) < 0.$$

The prediction is a larger increase in the explicit subsidy among exporters ($N_i > 0$) than among importers, and a larger increase the larger the net oil exposure. The first part motivates the binary design used for the main estimate; the second motivates the continuous-exposure variant reported in Section 7.

This is a heuristic, not a structural model. It ignores, among other things, the political economy of price adjustment and differences in Q_i and in the fuel mix. The most important omission is that pass-through also depends on the price-regulation regime, which differs across countries and is correlated with exporter status: a country with an administered price and a state-owned oil company may keep $\bar{p}_i$ fixed for reasons unrelated to its oil rents. The common response also varies with fuel intensity, Q_i relative to GDP, so the year effects absorb only its average; if exporters or high-subsidy countries are more fuel-intensive, part of the differential reflects intensity rather than rents. The design cannot separate the rent channel from such differences in regimes and intensity, and the estimates below are a differential response by exporter status, not a test of the rent mechanism.

4.2 Why difference-in-differences

The selection of method follows from the question and from the structure of the data. A before-and-after comparison of exporters does not isolate the shock, because other global elements also changed between 2021 and 2022, including the post-COVID-19 rebound and general inflation, and such a comparison would attribute all of them to the oil price. A comparison of exporters and importers in 2022 alone does not isolate it either, because the groups already differed before the shock: over 2015–2021, the average exporter's explicit subsidy was 3.74% of GDP, compared with 0.55% for the average importer. The data, however, have

two dimensions, time and group. Net importers went through the same period and faced the same price increase, but without the oil-rent channel. Comparing the change in exporters with the change in importers is a difference-in-differences comparison (Angrist and Pischke 2009). Because importers are also exposed, the comparison does not recover what exporters’ subsidies would have been without the shock; it recovers how much more they rose than those of a lower-exposure group with identical timing.

4.3 Specification

Let S_{it} denote the explicit subsidy of country i in year t , in percent of GDP. The baseline 2×2 regression is

$$S_{it} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Exporter}_i + \beta_3 (\text{Post}_t \times \text{Exporter}_i) + \varepsilon_{it},$$

where $\text{Post}_t = 1$ in 2022 and $\text{Exporter}_i = 1$ for the five net exporters. The main specification replaces the group and period terms with country and year fixed effects:

$$S_{it} = \beta_3 (\text{Post}_t \times \text{Exporter}_i) + \alpha_i + \lambda_t + \varepsilon_{it}.$$

The country effects α_i absorb everything that is constant within a country, including exporter status and the level of its subsidy regime. The year effects λ_t absorb the part of common shocks, such as the level of the Brent price, the pandemic, and the regional business cycle, that affects all countries equally; any response to them that differs systematically between exporters and importers is not absorbed and loads on β_3 . The coefficient β_3 measures the change in exporters’ subsidy between the 2015–2021 average and 2022, in excess of the corresponding change among importers. Because importers are also exposed to the price increase, β_3 is the difference between the shock’s effect on exporters and its effect on importers; the latter is absorbed by the 2022 year effect and is not identified.

The outcome is measured in percentage points of GDP, and parallel trends is assumed on that scale. Because exporters start from a much higher level, a proportional version of the assumption would imply a different differential; and because exporters’ GDP also rises with the oil price, part of any response in the numerator is offset in the ratio.

The static regressions use 2015–2022: seven pre-shock years and the shock year. The year 2023 is excluded from the static estimator because Brent had already fallen and pooling it with 2022 would average a shock year with a reversal year. It is kept in the event study, which shows whether the differential reverses. Each country receives equal weight. Standard errors are clustered by country, which allows arbitrary serial correlation within countries. With five treated clusters out of 34, however, conventional cluster-robust standard errors tend to be too small, and the reported p -values should be read as lower bounds (Cameron et al. 2008; MacKinnon and Webb 2017). Methods designed for this case, namely the wild cluster restricted bootstrap, randomization inference that reassigns exporter status across the 34 countries, and the procedure of Conley and Taber (2011), are the appropriate next step; I expect them to widen rather than narrow the confidence interval.

For the event study, I replace the single interaction with one interaction per year,

$$S_{it} = \sum_{k \neq 2021} \delta_k \mathbf{1}[t = k] \times \text{Exporter}_i + \alpha_i + \lambda_t + \varepsilon_{it},$$

estimated on 2015–2023, with 2021, the last pre-shock year, as the reference. The coefficients $\delta_{2015}, \dots, \delta_{2020}$ are six leads that trace the pre-shock gap between groups; δ_{2022} is the shock-year differential and δ_{2023} the differential one year later.

5 Results

5.1 The 2×2 comparison

The DiD estimate can be read directly from four group means (unweighted country-years). Among net exporters, the explicit subsidy averaged 3.74% of GDP in 2015–2021 and 5.70% in 2022, an increase of 1.95 pp. Among net importers, it went from 0.55% to 1.18%, an increase of 0.62 pp. Importers’ subsidies therefore also rose, which is why the double difference, +1.33 pp of GDP, is a differential rather than a total effect. The four cells contain 35, 5, 200 and 29 observations; the total of 269 reflects the missing Puerto Rico observations noted in Section 2.

5.2 Two-way fixed effects estimates

Table 3 reports the regression estimates. Column (1) reproduces the 2×2 comparison as a regression: $\hat{\beta}_3 = 1.33$ (SE 1.03, $p = 0.20$). Column (2), the main specification with country and year fixed effects, gives 1.32 pp of GDP (SE 1.03, $p = 0.21$, 95% CI $[-0.79, 3.42]$). The two estimates nearly coincide, as they must in a design with a single treatment date and a nearly balanced panel; column (1) shows where the TWFE coefficient comes from, and their agreement is not evidence of robustness. The point estimate is economically meaningful, about 35% of the average pre-shock subsidy of exporters, but it is not statistically significant at conventional levels, and its 95% confidence interval includes both zero and differentials more than twice as large as the point estimate. With five treated countries, and with the few-cluster problem discussed in Section 4, this p -value is more likely to be too small than too large. This is the main, pre-specified result, and I report it as such rather than searching for a classification or specification that yields a smaller p -value; Section 7 reports the alternatives.

5.3 Auxiliary outcomes: implicit and total subsidy

If the estimated differential reflects the price channel, it should appear in the explicit subsidy, which is a price gap, more than in the implicit subsidy, which is dominated by structural components. Column (3) estimates the same TWFE model with the implicit subsidy as the dependent variable. The coefficient is -0.09 pp of GDP (SE 0.41, $p = 0.83$, 95% CI $[-0.93, 0.75]$): negative and far from significant. The implicit subsidy is not independent of the explicit one, however: it is measured relative to the higher of the retail price and the supply cost and includes forgone VAT, so a price shock can move it mechanically in either direction. The null

Table 3. Differential response of subsidies to the 2022 shock, net oil exporters relative to net oil importers: difference-in-differences estimates

Variable	(1) Explicit 2×2 DiD	(2) Explicit TWFE	(3) Implicit TWFE	(4) Total TWFE
Post2022 × Net exporter	1.331 (1.025)	1.319 (1.035)	−0.091 (0.412)	1.301 (0.982)
Post2022	0.624* (0.260)			
Net exporter	3.188 (1.994)			
Constant	0.553*** (0.147)			
Country FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
N (country-years)	269	269	228	228

Notes: β_3 (Post2022 × Net exporter) is the change in the subsidy of net oil exporters between the 2015–2021 average and 2022 in excess of the corresponding change among net oil importers (the comparison group). Dependent variable (% of GDP): explicit subsidy in columns (1)–(2), implicit in (3), and total in (4). Columns (2)–(4) include country and year fixed effects. For the implicit component, the effect is −0.09 pp of GDP and not statistically significant. Standard errors are clustered by country in parentheses. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Sample: 2015–2022 (34 countries; 30 in columns (3)–(4)). 2023, a reversal year (Brent fell from USD 100.9 to USD 82.5 per barrel), is excluded. *Source:* IMF Fossil Fuel Subsidies Database; U.S. Energy Information Administration (EIA).

result is therefore consistent with, but does not isolate, a price channel, and its confidence interval admits differential effects of up to 0.75 pp. Because the explicit differential is itself not significant, the contrast between the two outcomes is weaker evidence than it would otherwise be. The sample also differs: the falsification regression has 30 countries and 228 observations because the implicit subsidy is missing for Aruba, Puerto Rico, St. Kitts and Nevis, and Suriname and partly for Antigua and Barbuda and Dominica, so columns (2) and (3) are not estimated on the same countries. Column (4) uses the total subsidy: 1.30 pp (SE 0.98, $p = 0.20$). It is close in magnitude to the explicit differential and, like it, not statistically significant; because the samples differ and the IMF total is computed separately from its components, columns (2)–(4) need not add up.

6 Identification checks

6.1 Parallel trends

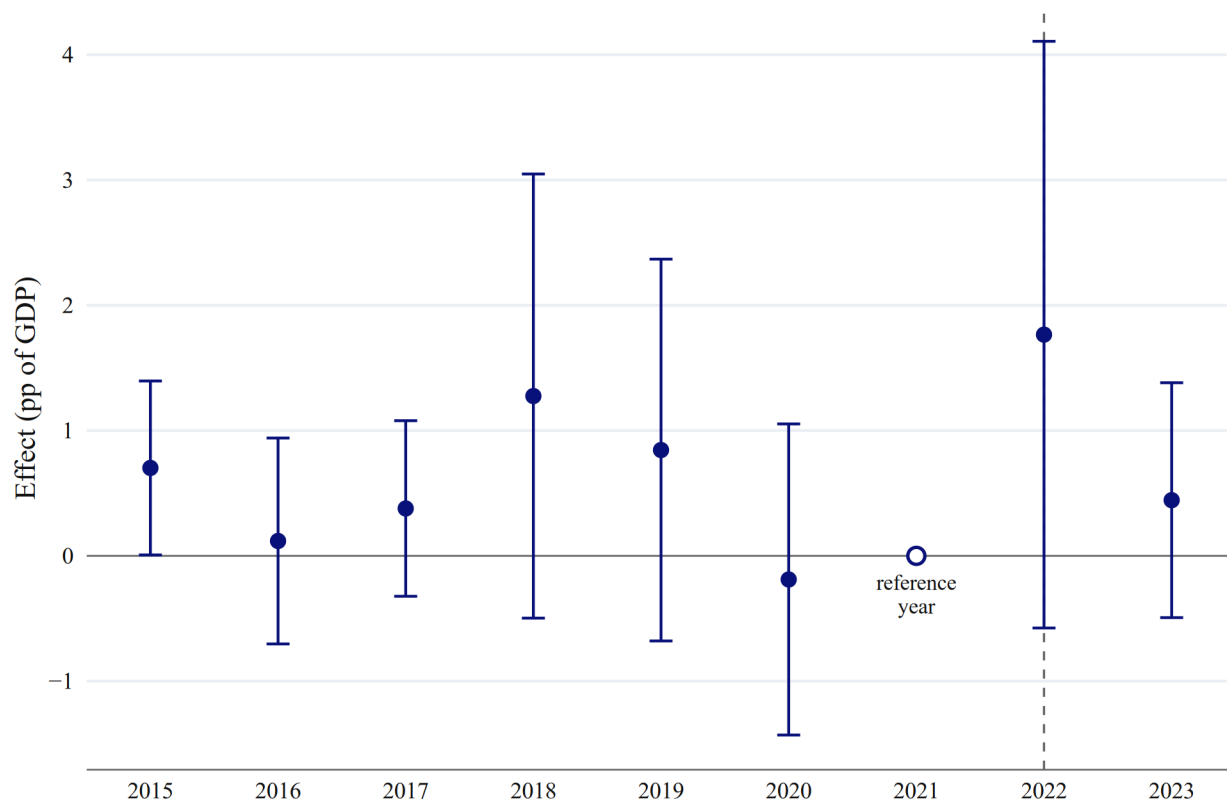
The identifying assumption of the static estimator is that, absent the difference in exposure, the explicit subsidy of exporters and importers would have changed by the same amount between the pre-shock average and 2022:

$$\mathbb{E} \left[S_{i,2022}^0 - \bar{S}_{i,\text{pre}}^0 \mid \text{Exporter}_i = 1 \right] = \mathbb{E} \left[S_{i,2022}^0 - \bar{S}_{i,\text{pre}}^0 \mid \text{Exporter}_i = 0 \right],$$

where $\bar{S}_{i,\text{pre}}^0$ is the 2015–2021 average, and S^0 is the subsidy that exporters would have had with an importer-type exposure. The event study uses the same condition with 2021 in place of the pre-shock average. The

assumption cannot be tested directly, but the event study shows whether the two groups were already separating before 2022. Figure 4 plots the coefficients.

Figure 4. Event study: differential response of the explicit subsidy by year (base year 2021)



Notes: Coefficients on the year $\times$ net oil exporter interactions (percentage points of GDP), with 2021 as the reference year (hollow point), and 95% confidence intervals from standard errors clustered by country. The dashed line marks the 2022 shock. Before 2022, the coefficients fluctuate without a monotonic trend; a joint test does not reject that all six are zero ($F(6, 33) = 1.37$, $p = 0.257$). Groups: five net oil exporters and 29 net oil importers. $N = 303$ country-years (34 countries, 2015–2023; Puerto Rico 2015–2017 missing). *Source:* IMF Fossil Fuel Subsidies Database.

The six pre-shock coefficients are +0.70 (2015), +0.12 (2016), +0.38 (2017), +1.28 (2018), +0.84 (2019), and -0.19 (2020). Five of their 95% confidence intervals include zero; the exception is 2015, whose lower bound is barely above zero (0.01). The largest coefficient appears in 2022 ($\hat{\delta}_{2022} = +1.77$, 95% CI [-0.58, 4.11]), and it declines in 2023 ($\hat{\delta}_{2023} = +0.44$, 95% CI [-0.49, 1.38]), in line with the fall in oil prices. Neither interval excludes zero.

A joint Wald test does not reject the null that the six leads are all zero: $F(6, 33) = 1.37$, $p = 0.26$, with the denominator degrees of freedom set to the number of clusters minus one. This is weak support for parallel trends, not a validation. With five treated clusters and lead standard errors of up to 0.87 pp, the test has little power against pre-trends of the size of the estimated effect, and conditioning the analysis on passing such a pre-test can itself distort estimates and inference (Roth 2022). With a single post-period, β_3 also cannot separate the response to the shock from any other exporter-specific shock in 2022: a group-specific deviation

of the size seen before 2022 would enter the point estimate directly rather than only widening its confidence interval (Conley and Taber 2011).

The pattern of the leads matters for interpretation. Five of the six are positive; only 2020 is negative. Relative to 2021, the exporter–importer gap was therefore larger in most earlier years, and the reference year is a comparatively low point for exporters. In the aggregates of Table 1, importers’ subsidies rebounded more than exporters’ between 2020 and 2021. This is why $\hat{\delta}_{2022}$, measured against 2021, is larger than the static $\hat{\beta}_3$ (1.32 pp), which is measured against the 2015–2021 average. The leads also co-move with the Brent price: over 2015–2020, their correlation with the annual Brent average is 0.95, and the largest lead, +1.28 in 2018, falls in the pre-shock year with the highest price among the lead years. The reference year does not fit this pattern: Brent in 2021 (USD 70.9) was close to its 2018 level (USD 71.3), yet the gap was, by construction, zero, and including the 2021 point lowers the correlation to 0.57. Measured against 2018 rather than 2021, the 2022 differential would be +0.49 pp. The leads are thus consistent with exporters’ subsidies responding somewhat more to the oil price every year, in which case the pre-period is not an untreated baseline, but seven points with one discordant year cannot establish such a pattern. I read β_3 as the differential response to an unusually large price increase, estimated with considerable uncertainty and sensitive to the choice of the pre-shock benchmark.

A formal sensitivity analysis that bounds post-period violations of parallel trends by the size of pre-period violations (Rambachan and Roth 2023) is a natural extension. Because the 95% interval for δ_{2022} already includes zero, such an analysis could only widen it.

6.2 No anticipation

The second premise is that exporters’ subsidies in 2021 did not respond to a shock that had not yet occurred. With a sole shock year, this assumption cannot be tested with the data, and I do not claim to have validated it. The argument is institutional: the invasion of Ukraine in February 2022 was unanticipated, and there is no indication that any government in the region repositioned its fuel subsidies in 2021 in expectation of a price increase in 2022. The leads show no sign of an anticipatory rise among exporters: in 2021 the exporter–importer gap was smaller than in five of the six earlier years. The more relevant concern is that prices were already rising in 2021, so the base year is partly exposed. To the extent that the 2021 differential response has the same sign as the 2022 one, this biases the 2022 coefficient toward zero rather than away from it.

6.3 No spillovers

The design also assumes that one country’s treatment status does not affect subsidies in another, the stable unit treatment value assumption. Regional energy trade could violate it. Venezuela and Trinidad and Tobago supply crude oil and refined products to Caribbean neighbours, and Colombia and Ecuador trade fuels with their neighbours, so the terms on which exporters supply the region can shift the supply cost, and hence the explicit subsidy, of importers. Flows can also run the other way: Bolivia, a net oil importer under the main rule, sells natural gas to Argentina and to Brazil, a treated country, so a comparison country’s windfall can affect a treated one, which could bias β_3 in either direction. Spillovers that affect all countries equally are

absorbed by the year effects; only those that fall differently on the two groups bias β_3 . If exporters shared part of their 2022 windfall with importing neighbours through cheaper supply, importers' subsidies would rise less than they otherwise would, and β_3 would overstate the differential. If exporters instead cut supply to neighbours when prices rose, importers' costs would rise more, and β_3 would understate it. The direction of the bias is therefore ambiguous, and with country-level annual data, the assumption cannot be tested here.

7 Robustness & inference

7.1 Extreme cases and leave-one-out

Two countries are extreme in the distribution of the outcome. Venezuela's explicit subsidy reached 18.1% of GDP in 2022, the highest among exporters, and Suriname's reached 8.5%, the highest among importers. Dropping Venezuela from the main specification would amount to selecting on the outcome, so I keep it and instead report the main TWFE specification on alternative samples in Table 4. Without Venezuela, the estimate falls from 1.32 to 0.28 (SE 0.45, $p = 0.54$, 95% CI $[-0.63, 1.19]$). This is the most consequential sensitivity in the paper: most of the point estimate comes from one country, which is also the country whose GDP, the denominator of the outcome, is most uncertain after hyperinflation. Excluding Suriname raises the estimate to 1.48 (SE 1.02, $p = 0.16$), because it removes a large-subsidy importer from the comparison group. Excluding both yields 0.44 (SE 0.42, $p = 0.30$).

Table 4. Sensitivity of β_3 to the exclusion of extreme cases

Variable	(1) Full sample	(2) Without Venezuela	(3) Without Suriname	(4) Without both
Post2022 $\times$ Net exporter	1.319 (1.035)	0.279 (0.447)	1.484 (1.024)	0.445 (0.418)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N (country-years)	269	261	261	253

Notes: Sensitivity of β_3 (Post2022 $\times$ Net exporter) to excluding the countries with the most extreme subsidies. All columns use the TWFE specification of column (2) of Table 3; only the sample changes. Column (1) is the full sample (34 countries). Columns (2)–(4) exclude Venezuela (the net oil exporter with the highest subsidy, 18.1% of GDP), Suriname (the net oil importer with the highest subsidy, 8.5% of GDP), and both. Excluding each of the five net oil exporters in turn gives estimates between 0.28 (without Venezuela) and 1.81 (without Brazil); this is a range of point estimates, not a confidence interval. Standard errors are clustered by country in parentheses. $\dagger p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$. *Source:* IMF Fossil Fuel Subsidies Database.

A systematic version of the same exercise drops each of the five exporters in turn. The coefficient remains positive in every case, ranging from 0.28 (without Venezuela) to 1.81 (without Brazil), and none of the five estimates is significant at the 10% level. Dropping Brazil raises the estimate because its explicit subsidy is close to zero throughout and fell marginally in 2022. These re-estimates have only four treated clusters and inherit the inference problem discussed below. The sign of the differential does not hinge on a single country; its size does, and it hinges on Venezuela. The exercise drops only treated units; dropping comparison

countries one at a time would show whether the composition of the comparison group matters, and is a further extension.

7.2 Sensitivity to the definition of exporters

Table 5 re-estimates the main specification under alternative definitions of exporters. Panel A reports the variants that were specified together with the main rule, before estimation; Panel B reports three variants that I added after inspecting the classification and the continuous-exposure result, and they should be read as post hoc. Row (1) reproduces the main estimate.

Table 5. Sensitivity of β_3 to the definition of net oil exporters

Classification variant	Coefficient	SE	95% CI	p-value	N	Treated
Panel A. Pre-specified variants						
(1) Main rule (HS 2709+2710, 2015–2019)	1.319	(1.035)	[−0.79, 3.42]	0.211	269	5
(2) Including gas (HS 2711)	2.060 $\hat{a}$	(1.013)	[−0.00, 4.12]	0.050	269	6
(3) 2019–2021 average	0.916	(0.944)	[−1.00, 2.84]	0.339	269	6
(4) Main rule, without Guyana	1.286	(1.038)	[−0.83, 3.40]	0.224	261	5
(5) Previous hand-coded list	1.790 $\hat{a}$	(0.920)	[−0.08, 3.66]	0.060	269	7
(6) Continuous exposure (per 1 pp of GDP)	0.223***	(0.030)	[0.16, 0.28]	<0.001	264	$\hat{a}$
Panel B. Post hoc variants (not pre-specified)						
(7) Main rule, without Puerto Rico	1.300	(1.038)	[−0.81, 3.41]	0.220	264	5
(8) Main rule, without Trinidad and Tobago	1.623	(1.230)	[−0.88, 4.13]	0.196	261	4
(9) Continuous exposure, without Venezuela	0.130**	(0.047)	[0.03, 0.23]	0.010	256	$\hat{a}$

Notes: Each row re-estimates $\hat{\beta}_3$ (Post2022 – Net exporter) with the specification of column (2) of Table 3 (explicit subsidy, % of GDP; country and year fixed effects; 2015–2022); only the classification or the sample changes. Main rule (1): net oil exporter if average 2015–2019 exports minus imports of crude oil and refined products (HS 2709 + 2710, UN Comtrade, partner World; mirror data for countries that do not report) are positive; this gives five net oil exporters (Brazil, Colombia, Ecuador, Trinidad and Tobago, and Venezuela). Relative to it, (2) adds petroleum gases (HS 2711) and moves Bolivia to net exporter; (3) averages 2019–2021 and moves Guyana to net exporter; (4) drops Guyana (a net importer under the main rule) from the sample; (5) uses the previous hand-coded list and moves Bolivia, Guyana, and Mexico to net exporter and Brazil to net importer. (6) replaces the binary treatment with Post2022 – net oil trade as a percentage of GDP (2015–2019 averages); its coefficient is the change in the explicit subsidy (pp of GDP) per 1 pp of GDP of net oil exports, so it is not comparable in size with the other rows. It excludes Puerto Rico, which has no Comtrade data and is classified as a net oil importer in every binary variant from U.S. Energy Information Administration data. Panel B was added after inspecting the classification (rows 7 and 8) and the result of row 6 (row 9) and was not pre-specified: (7) drops Puerto Rico (classified from an outside source); (8) drops Trinidad and Tobago, whose 2015–2019 average net oil trade (USD 8 million a year) is close to zero and changes sign across years; (9) re-estimates (6) without Venezuela, whose net oil exports (19.0% of GDP) far exceed the next largest (Ecuador, 4.2% of GDP). Standard errors clustered by country; p-values and 95% confidence intervals from the clustered standard errors. $\hat{a}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Source:* IMF Fossil Fuel Subsidies Database; UN Comtrade; U.S. Energy Information Administration.

Every binary definition gives a positive point estimate, between 0.92 and 2.06, and none is significant at the 5% level. Adding petroleum gases (row 2) moves Bolivia to the exporter group and gives the largest binary estimate, 2.06 ($p = 0.05$, with a 95% CI whose lower end is essentially zero). Bolivia’s classification matters because its subsidy rose by 5.38 pp in 2022 relative to its 2015–2021 average, the second-largest increase in the sample; as a gas exporter, it also received a windfall that the main rule does not count. Averaging over 2019–2021 (row 3) moves Guyana to the exporter group and gives 0.92 ($p = 0.34$); dropping Guyana (row 4) leaves the main estimate almost unchanged (1.29). The hand-coded list used in the earlier version of this analysis (row 5) gives 1.79 ($p = 0.06$). I report it only as a sensitivity check: the rule in row (1) was

fixed before estimation, and choosing among classifications after seeing their p -values would invalidate the inference. Dropping Puerto Rico (row 7) or Trinidad and Tobago (row 8), the two cases that rest on an outside source or a near-zero average, leaves the estimate positive and not significant (1.30 and 1.62). The sign of the differential is robust to how exporters are defined; its statistical significance is not.

Row (6) replaces the binary indicator with net oil trade as a percentage of GDP (2015–2019 average), interacted with Post_t . Each additional percentage point of GDP in net oil exports is associated with a 0.223 pp larger increase in the explicit subsidy in 2022 (SE 0.030, $p < 0.001$). This is the dose–response pattern predicted in Section 4.1: the larger the net oil exposure, the larger the differential response. I treat it as complementary evidence, not as the main result, for three reasons. First, the exposure distribution is highly skewed. Net oil trade ranges from -7.3% of GDP (Guyana) to 19.0% (Venezuela), with a median of -2.3% , and 28 of the 33 countries have negative values; Venezuela’s exposure is more than four times that of the next country, Ecuador (4.2%). The slope is therefore heavily leveraged by one country: without Venezuela (row 9, post hoc), it falls to 0.130 (SE 0.047, $p = 0.010$), 42% lower. Second, with 33 clusters and one high-leverage cluster, clustered standard errors are likely too small (MacKinnon and Webb 2017), so the p -value overstates the precision. Third, the continuous variant was pre-specified as a sensitivity check, not as the main specification, and promoting it after seeing that it is significant would be the kind of specification search the pre-specified rule is meant to prevent. I read it as suggestive evidence of a dose–response pattern that survives, in attenuated form, the exclusion of Venezuela.

7.3 Inference with few treated clusters

The small treated group is the main limitation of the design, and it affects inference in two ways. First, with five treated clusters, conventional cluster-robust standard errors understate variability because the variance of $\hat{\beta}_3$ is driven by the few treated clusters (Cameron et al. 2008; MacKinnon and Webb 2017), so the p -values in Tables 3, 4, and 5 are likely optimistic. This reinforces, rather than overturns, the conclusion that the binary estimates are not statistically significant. Second, idiosyncratic shocks to five countries do not average out: the estimator does not concentrate around the true differential as the number of comparison countries grows (Conley and Taber 2011), so any single estimate can be far from the true value even if it is correct on average. The dependence on Venezuela is a concrete instance of this problem. The wild cluster restricted bootstrap, randomization inference over exporter status, and the Conley and Taber (2011) procedure address these problems and are the natural next step; I have not run them, and I expect them to widen the confidence intervals. The recent econometric literature on DiD also discusses the interpretation of pre-trend tests and the problems of TWFE in the context of staggered adoption (Roth et al. 2023). The last concern does not arise here, because treatment starts in the same year for every treated country, so the TWFE estimator reduces to a comparison between one high-exposure cohort and a lower-exposure comparison group with identical timing.

8 Budgetary implications and policy

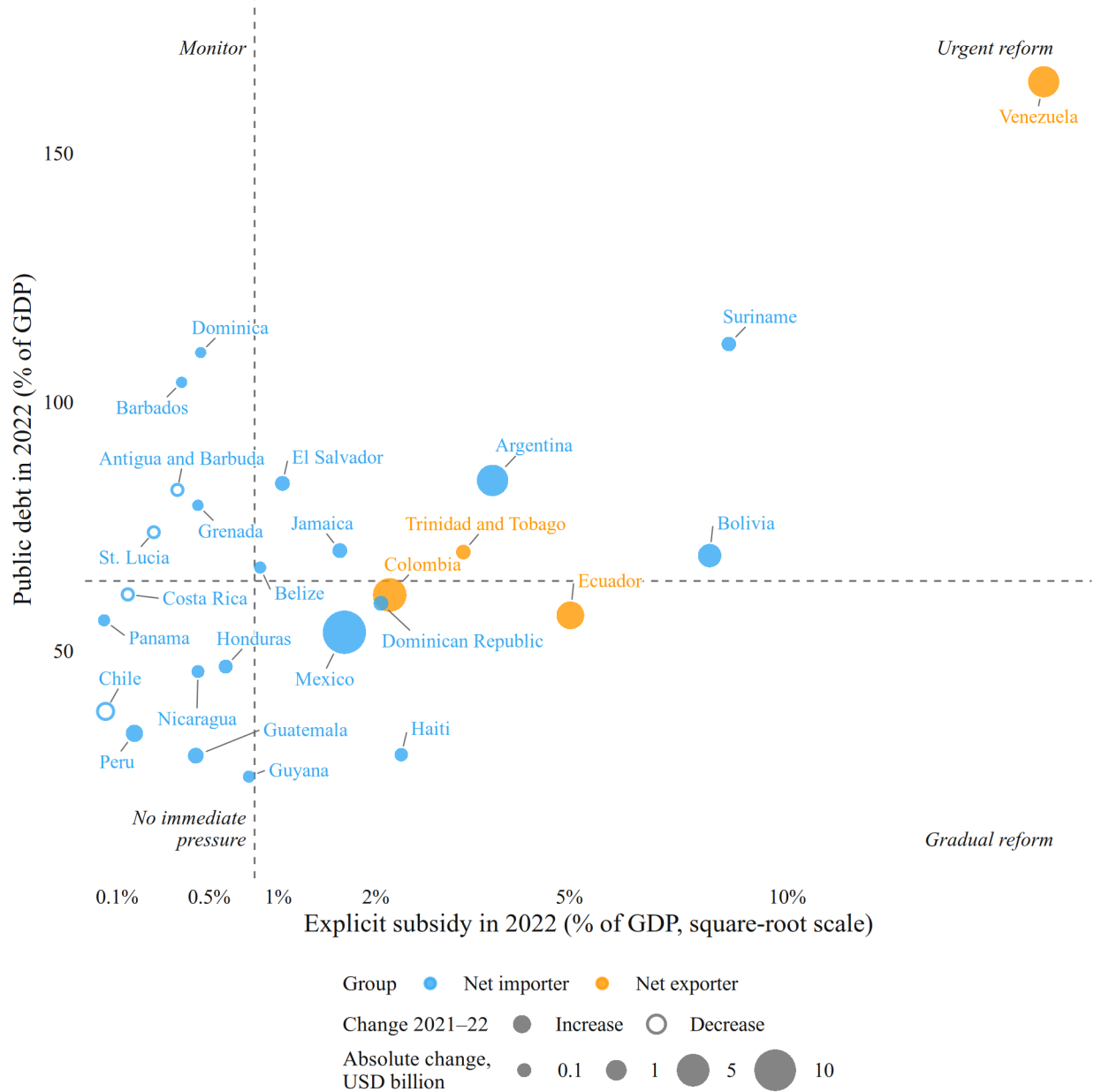
8.1 A subsidy–debt matrix

The estimate in Section 5 is an average differential; the policy question is where the subsidy is most costly relative to the fiscal room to sustain it. I build a descriptive matrix of the 26 countries with an explicit subsidy exceeding 0.05% of GDP in 2022 (4 exporters and 22 importers). Brazil, the remaining exporter, is not in the matrix: its explicit subsidy was 0.04% of GDP in 2022, below the floor. The horizontal axis shows the explicit subsidy in 2022, and the vertical axis shows gross public debt in 2022, both expressed as a percent of GDP. The sample medians, 0.81% of GDP for the subsidy and 64.1% of GDP for debt, divide the plane into four quadrants. I use medians rather than means because they are not driven by extreme values such as Venezuela. Fiscal space is proxied by the debt stock rather than the one-year deficit, which is volatile and cyclical. The matrix is a prioritization device, not a causal model: it indicates where reform is most pressing, not the effect of any particular policy. Figure 5 shows the matrix and Table 6 reports the underlying values.

The *urgent reform* quadrant (subsidy and debt both above the median) contains eight countries: Venezuela, Suriname, Bolivia, Argentina, Trinidad and Tobago, Jamaica, El Salvador, and Belize. It mixes two exporters (Venezuela and Trinidad and Tobago) with six importers (Suriname, Bolivia, Argentina, Jamaica, El Salvador, and Belize). Venezuela combines a subsidy of 18.1% of GDP with debt of 164.4% of GDP, Suriname 8.5% with 111.7%, and Bolivia 8.0% with 69.2%. Six of the eight also ran a fiscal deficit in 2022, the largest in Bolivia (−6.1% of GDP) and Venezuela (−5.3%). The *gradual reform* quadrant (high subsidy, debt below the median) contains Ecuador, Haiti, Colombia, the Dominican Republic, and Mexico, again a mix of exporters and importers. The *monitor* quadrant (low subsidy, high debt) contains Dominica, Grenada, Barbados, Antigua and Barbuda, and St. Lucia, all small Caribbean economies with limited fiscal room. The *no immediate pressure* quadrant contains Guyana, Honduras, Nicaragua, Guatemala, Peru, Costa Rica, Chile, and Panama. The cut-offs are sample medians, and some countries sit close to them: Belize, Colombia, Costa Rica, the Dominican Republic, and Guyana are within 5 points of GDP of the debt median or within 0.1 pp of the subsidy median, and could change quadrant under small changes in the thresholds.

As an illustration only, Table 6 values the average differential of 1.32 pp of GDP at the 2022 GDP of each of the four exporters in the matrix. The figure is how much more an exporter of that size would have spent than an importer-type response implies, *if* it responded like the average exporter. It is neither the total cost of the shock nor the country’s realized cost, and it inherits the uncertainty of $\hat{\beta}_3$, whose 95% CI of [−0.79, 3.42] includes zero, so the lower end of every range is negative. For Colombia, the largest of these economies, the valuation is USD 5.0 billion, with a range of USD −3.0 billion to USD 13.1 billion across the confidence interval, compared with a realized 2021–2022 increase of USD 5.4 billion; for Ecuador it is USD 1.8 billion (range −1.0 to 4.5), and for Venezuela, whose GDP is the most uncertain, USD 0.75 billion. No figure is reported for importers: they are the comparison group, so their differential is zero by construction, and the model identifies nothing about the shock’s cost for them, which is absorbed by the year effects. The observed change is shown in a separate column (in pp); its absolute value in USD billion sets the size of the points in Figure 5, with hollow points for decreases.

Figure 5. Subsidy–debt matrix and policy quadrants, 2022



Notes: Countries with an explicit subsidy above 0.05% of GDP in 2022 (26: four net oil exporters and 22 net oil importers). Horizontal axis on a square-root scale. Point size is the absolute 2021–2022 change in the explicit subsidy (USD billion); hollow points are decreases. Dashed lines: sample medians of the subsidy (0.81% of GDP) and of gross public debt (64.1% of GDP). The estimated differential effect for net oil exporters is 1.32 pp of GDP (Table 3), between 0.28 and 1.81 when each exporter is excluded in turn (Section 7). Net oil importers such as Suriname and Bolivia are also in the urgent reform quadrant. Debt and the fiscal balance are not included in the estimation. *Source:* IMF Fossil Fuel Subsidies Database; IMF, World Economic Outlook.

Table 6. Subsidy, illustrative differential cost and fiscal space by country, 2022

Country	Subsidy (% of GDP)	Change 2021–22 (pp)	Illustrative cost (USD billion)	Debt (% of GDP)	Balance (% of GDP)	Quadrant
Panel A. Net oil exporters						
Venezuela	18.06	+6.81	0.75	164.4	–5.3	Urgent reform
Ecuador	5.02	+2.07	1.75	57.2	0.0	Gradual reform
Trinidad and Tobago	3.19	+0.53	0.33	69.9	+1.0	Urgent reform
Colombia	2.17	+1.36	5.04	61.3	–6.4	Gradual reform
N (countries)	4					
Panel B. Net oil importers						
Suriname	8.49	+3.78	n.a.	111.7	–2.6	Urgent reform
Bolivia	8.02	+3.22	n.a.	69.2	–6.1	Urgent reform
Argentina	3.65	+0.70	n.a.	84.3	–3.8	Urgent reform
Haiti	2.32	+0.43	n.a.	29.2	–1.8	Gradual reform
Dominican Republic	2.06	+0.05	n.a.	59.6	–3.2	Gradual reform
Mexico	1.64	+0.72	n.a.	53.8	–4.3	Gradual reform
Jamaica	1.59	+0.84	n.a.	70.2	+0.3	Urgent reform
El Salvador	1.04	+0.46	n.a.	83.7	–2.7	Urgent reform
Belize	0.85	+0.43	n.a.	66.8	–1.0	Urgent reform
Guyana	0.77	–0.28	n.a.	24.8	–5.1	No immediate pressure
Honduras	0.60	+0.27	n.a.	46.9	+1.6	No immediate pressure
Dominica	0.45	–0.01	n.a.	110.0	–7.2	Monitor
Nicaragua	0.43	+0.23	n.a.	45.9	+0.6	No immediate pressure
Grenada	0.43	+0.10	n.a.	79.3	+0.9	Monitor
Guatemala	0.42	+0.23	n.a.	29.0	–1.7	No immediate pressure
Barbados	0.35	–0.03	n.a.	104.0	–1.7	Monitor
Antigua and Barbuda	0.33	–0.04	n.a.	82.4	–2.9	Monitor
St. Lucia	0.22	–0.04	n.a.	73.9	–2.5	Monitor
Peru	0.15	+0.15	n.a.	33.5	–1.4	No immediate pressure
Costa Rica	0.13	–0.01	n.a.	61.4	–2.7	No immediate pressure
Chile	0.07	–0.08	n.a.	37.9	+1.4	No immediate pressure
Panama	0.07	+0.01	n.a.	56.2	–4.0	No immediate pressure
N (countries)	22					

Notes: Countries with an explicit subsidy above 0.05% of GDP in 2022, sorted by subsidy within each group. Change 2021–22: observed change in the explicit subsidy (pp of GDP), descriptive. Illustrative cost: β_3 (1.32 pp of GDP, Table 3) times each net oil exporter's 2022 GDP (USD billion), an illustrative differential cost (exporters' subsidy increase beyond importers'), not the total cost of the shock; the 95% confidence interval of β_3 (–0.79 to 3.42) includes zero. n.a.: net oil importers, the comparison group. Debt: gross public debt; Balance: general government fiscal balance (negative = deficit); both % of GDP. Quadrant: position relative to the sample medians of the subsidy (0.81% of GDP) and debt (64.1% of GDP). *Source:* IMF Fossil Fuel Subsidies Database; IMF, World Economic Outlook.

The main lesson of the matrix is that fiscal pressure from fuel subsidies does not align with the oil trade position. Whatever the size of the average differential, importers such as Suriname, Bolivia, and Argentina are among the most exposed. High subsidies combined with high debt were present before 2022 and, in the raw data, rose further in several of these countries in 2022 (Table 2); the model does not identify how much of that rise in importers is due to the shock.

8.2 Regressivity and the case for targeted compensation

Removing a subsidy raises fuel prices and lowers households' real income, so the policy question is not whether to reform but how to protect low-income households during the transition. Universal fuel subsidies are a poorly targeted instrument. Because benefits are proportional to consumption, which rises with income, they accrue mostly to richer households: for developing countries, Coady et al. (2015) find that the richest income quintile captures more than 40% of the benefits of fuel subsidies, about six times the share of the poorest quintile. Replacing universal price subsidies with targeted transfers to vulnerable households can improve equity and free up fiscal space by ending transfers to higher-income households. A well-designed reform need not be regressive.

8.3 A differentiated reform sequence

The recommendation varies by quadrant. In the urgent reform quadrant, domestic price adjustment should come first, accompanied by immediate targeted compensation; several of these countries already run deficits, and there is little room to postpone the adjustment. In the gradual reform quadrant, lower debt allows a phased transition with an announced schedule and compensation for vulnerable households. In the monitor and no-immediate-pressure quadrants, the priority is surveillance: because the explicit subsidy rises with oil prices, a new price shock can move these countries toward the urgent reform quadrant. Across all groups, a common sequence applies, compressed in time for the urgent reform quadrant: first, adopt a pricing rule that passes international price changes through to domestic prices, with smoothing if needed; second, close the remaining price gap on an announced path; third, redirect the savings to targeted transfers.

9 Conclusion

Under a treatment rule fixed before estimation, the explicit fossil fuel subsidy of LAC net oil exporters rose by about 1.32 pp of GDP more than that of net importers in 2022. The estimate is imprecise and not statistically significant ($p = 0.21$; 95% CI $[-0.79, 3.42]$), and with five treated countries, conventional clustered standard errors are likely too small, so the true uncertainty is larger than reported. Importers were also exposed to the shock, so this is a differential response, not the total effect of the shock. The point estimate is positive under every binary definition of exporters considered (0.92 to 2.06 pp) and when each exporter is dropped in turn, so its sign is robust; its size is not, since it falls to 0.28 pp without Venezuela. A continuous measure of net oil exposure points in the same direction, with a larger differential response the larger a country's net oil exports, but it is heavily leveraged by Venezuela and was pre-specified only as a sensitivity check.

The timing is consistent with a price channel: the differential peaks in 2022 and falls in 2023 as oil prices decline, and the implicit subsidy shows no statistically significant differential response. The joint pre-trend test does not reject, but it has little power, and the leads co-move with the Brent price before 2021. The main caveats are the small treated group and the weight of one country; the confounding of exporter status with price-regulation regimes; possible spillovers through regional energy trade; and a falsification outcome that is not independent of the explicit measure. Overall, the evidence is suggestive of a larger subsidy response among net oil exporters, not conclusive.

These caveats point to concrete extensions: small-cluster and leverage-robust inference (wild cluster restricted bootstrap, randomization inference, and the Conley and Taber (2011) procedure), which matters most for the continuous-exposure estimate; a sensitivity analysis for violations of parallel trends (Rambachan and Roth 2023); a design that interacts net oil exposure with the Brent price in every year; and direct measures of price-regulation regimes, which would help separate the oil-rent channel from differences in pricing institutions.

The fiscal analysis shows that high subsidies combined with high debt are not limited to exporters. For importers, a subsidy tied to the international price expands exactly when supply is most expensive; for exporters, it absorbs the oil windfall that could otherwise be saved. In both cases, it is procyclical. Reforming it, in a sequence adapted to each country's fiscal space and with targeted compensation in place of universal price support, would free fiscal resources and reduce the exposure of public finances to oil price volatility.

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